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ORIGINAL INVESTIGATION

External Training Loads and Soft-Tissue Injury Occurrence During Congested Versus Noncongested Periods in Football

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Objective: To analyze the influence of congested and noncongested fixture periods during 2 seasons in a professional male football team on soft-tissue injury incidence and external load. ***Methods:*** Thirty-three professional football players from a Portuguese Liga I team participated in this study. Weekly external load and soft-tissue injury rate and burden of 2 consecutive seasons (2021–22 and 2022–23) were analyzed. ***Results:*** Total soft-tissue injury rate and burden for the 2 seasons were 3.9 and 3.2 injuries per 1000 hours and 71.8 and 60.5 days per 1000 hours for congested and noncongested periods, respectively. No significant differences were observed between congested and noncongested periods. Total high-speed running, sprint distance, distance above 80% and 90% of maximal velocity, and meters accelerating and decelerating above 2 m/s^2 were significantly higher for noncongested weeks. Match accelerations and decelerations above 3 m/s^2 were higher during congested periods and training during noncongested periods. No differences between the 2 periods were observed for the total number of accelerations and decelerations above 3 m/s^2. Overall, physical outputs per week were higher for training during noncongested weeks, whereas matches during congested periods registered higher external load. ***Conclusions:*** No effect of a congested schedule was observed on soft-tissue injury rates and injury burden. Higher match exposure during congested periods increased external load performed per week, and during noncongested periods, training load was superior to congested weeks.

Keywords: monitoring, mechanical load, musculoskeletal

In football, periods when official matches are interspersed by 96 hours or less have become a present reality.[1] These congested fixture periods can occur frequently throughout a football season, with players competing in both domestic and international competitions.[2,3]

Congested fixture periods have been associated with increased injury risk[4,5] and particularly soft tissue noncontact injuries.[3,6–8] Bengtsson et al[7] reported increases in muscle injuries in an 11-year follow-up of the Union of European Football Associations (UEFA) Champions League study, mainly for hamstring and quadriceps muscles, with a relative risk increase from 1.09 to 1.32 when <4 days separate 2 consecutive matches. The same group found no increases in total injury rates during match-congested periods in a subsequent analysis but confirmed a decreased incidence of muscle injuries when matches were played for a minimum of 6 days.[8] Hamstrings and quadriceps injuries were also increased within the total muscle injuries observed in the study of Carling et al[2] during 6 seasons in a French Ligue 1 team, with a higher incidence in the final 15 minutes of the second half for matches played in short, congested periods. More recently, Pinheiro et al[9] observed an injury incidence of 19.0 per 1000 hours of muscle injuries in a prolonged congested fixture period.

Insufficient recovery, traveling schedules, and increased competition exposure during congested periods are associated with physical and mental fatigue in football players,[4] and therefore are a potential cause for higher injury risk observed during these periods. The impact on injury incidence of external load from training and competition has been assessed through the analysis of physical outputs, often comparing congested versus noncongested periods.[8] The analysis of physical outputs also informs about the individual player's contribution to the team performance during different fixture periods. With this purpose, a study analyzing player's acceleration and deceleration throughout congested and noncongested periods has shown a decrease in congested periods, possibly reflecting the accumulation of fatigue and thus the inability to maintain a similar level of performance throughout consecutive matches.[9] A recent meta-analysis by Julian et al[10] showed decreases in certain physical outputs during congested periods, such as low- and moderate-distance running. According to the authors, the reduction in locomotive load might be attributed to pacing strategies used by players to allow the maintenance of higher speed distance zones. The importance of analyzing load parameters throughout a season in which congested periods alternate with regular ones might be relevant to increase the knowledge of potential injury incidence factors during these periods. For example, fluctuations of load parameters, respectively, to season averages were shown to be associated with noncontact soft tissue injuries, with higher meters per minute and a lower accumulation of an accelerometer-derived metric (ie, new body load) preceding injury occurrence.[11] Additionally, 1 to 4 weeks of cumulative external load has also been associated with injury risk.[12] Specifically, these authors showed that, during the preseason, the 3-week total distance and sprint distance were associated with a higher risk of a player sustaining an injury, whereas during in-season 3- and 4-week acceleration-related metrics presented similar results regarding injury risks. From a different perspective, adequate training chronic load and aerobic status seemed to be associated with a better tolerance to weekly large increases in speed distances (14.4 and 19.8 km/h) and a decreased risk of injury throughout a football season.[13]

To the best of our knowledge, only 2 studies have observed match and training external load during congested and noncongested

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weeks for entire seasons. Howle et al[5] found a decrease in total distance and several speed zone distances and meters per minute per week when matches were played within <4 days. Similarly, Garcia et al[14] observed that regular weeks presented a higher accumulation of external load than congested weeks for both accelerations/decelerations and speed zones above 18 km/h distances. In their study, particular attention was given to players involved regularly in >60 minutes of competitions during all matches for the congested period, with those being exposed generally to a higher workload than the remaining players both during congested and regular weeks. The increased overall load in regular weeks results from the greater frequency of high-load volume training sessions between matches. On the other hand, during short turnarounds, training days aim to recover and prime for matches. Importantly, these 2 studies used thresholds of 18 and 20.5 km/h, which should be interpreted with caution. Such thresholds do not isolate higher speed zone distances that usually reflect higher musculoskeletal mechanical strain associated with football running efforts.[15] The importance of higher-intensity efforts, as reflected by the accumulation of distance in speed zones closer to football players' maximum velocity, is related to the demands placed on the musculoskeletal system, particularly in lower limb muscles.

As a result of the association of external load and injury, the analysis of physical outputs and injury rates for training and competition during congested and noncongested periods throughout a season might be relevant. This analysis could inform about the potential impact that congested periods might have on noncontact soft tissue injuries. Additionally, including relative velocity distance zones above 80% and 90% of a player's maximum velocity, due to their mechanical strain potential, can potentially add a new element of information to the existing knowledge in this field. Therefore, the purpose of this study was to analyze the influence of congested and noncongested fixture periods during 2 entire seasons in a professional male football team on soft tissue injury and external load. Our hypothesis was 2-fold: (1) congested periods might increase the incidence of soft tissue injuries due to increased match exposure, and (2) fluctuations of match and training external load during congested and noncongested periods could influence soft tissue injury risk.

Methods

Participants

A total of 33 professional football players (age: 24.7 4.5] y; height: 189.3 [7.0] cm; body mass: 76.9 [6.4] kg) from a Portuguese Liga I team participated in this study. External load and soft tissue injury incidence for 2 consecutive seasons (2021–2022 and 2022–2023) were analyzed. There was a total of 22 and 26 squad players involved in training and/or competitions in the 2021–2022 and 2022–2023 seasons, respectively. From these, 15 players (age: 25.1 [4.5] y; height: 179.2 [7.1] cm; body mass: 77.1 [7.0] kg) were present in both seasons, 7 (age: 25.9 [5.1] y; height: 181.1 [7.4] cm; body mass: 78.1 [6.0] kg) only for the 2021–2022 season, and 11 (23.3 [3.6] y; height: 181.1 [6.5] cm; body mass: 75.9 [5.7] kg) for the 2022–2023 season. The squad comprised 20 outfield players during the 2 entire seasons, except for short periods of transfer windows.

The team was involved in a total of 134 matches, 105 official matches (n = 52, season 2021–2022; n = 53, season 2022–2023), and 29 friendly matches (n = 14, season 2021–2022; n = 15, season 2022–2023). The official matches included the Portuguese Liga I (n = 68), Portuguese Cup (n = 7), and Portuguese League Cup (n = 10) and from the UEFA. Matches for UEFA competitions included for both seasons group stage of the UEFA Champions League (n = 12) and playoffs 8 finals (n = 2) in 2021–2022, and UEFA Europa League last 16, 8, and quarterfinals (n = 6) in 2022–2023. A total of 432 training sessions were performed, 213 and 219 training for the 2021–2022 and 2022–2023 seasons.

Procedures

Congested fixture periods were considered when official matches were played within 96 hours or less.[10] For the 2 seasons, 15 congested fixture periods were registered, being 8 and 7 periods for the 2021–2022 and 2022–2023 seasons, respectively. These periods correspond to 225 days and 71 games, interspersed by 17 noncongested periods, in a total of 402 days, constituted by 9 and 8 periods for the 2021–2022 and 2022–2023 seasons, respectively.

External-Load Data Collection

External-load data from games and training were collected using 10-Hz Global Positioning System (GPS) units (APEX, STATSports) incorporated into the players' jerseys on the upper thoracic spine between the scapulae. To decrease the between-unit variability, the same GPS unit was used by each participant for subsequent sessions. The GPS units were turned on before the warm-up and turned off after the completion of the training sessions. After each training session, the raw data files were analyzed, and individual sessions' total high-speed running (HSR) distance and sprint distance, which reflected the distance covered in meters above 19.8 and 25 km/h, respectively, relative velocity distance above 80% (V80) and 90% (V90) of the individual maximum speed, the sum of distance accelerating and decelerating above 2 m/s^2 (AD2), and the sum of the number of accelerations and decelerations above 3 m/s^2 (AD3) were obtained from the company's software (STATSports Sonra software, version 4.0, STATSports). All variables were expressed in meters, except for AD3 that are reported in frequency.

Injury Data Collection

Soft tissue injuries included muscle and tendon medical attention and time-loss injuries, resulting from an acute indirect mechanism or with a gradual onset. Injury distribution and classification (Tables 1 and 2) were adopted using the football-specific extension of the International Olympic Committee consensus method of recording and reporting epidemiological data on injury and illness in sport.[16,17] Injuries were defined as any physical complaint resulting from a football match or training requiring medical attention. Medical attention injuries and time-loss soft tissue injuries are separated and detailed in Tables 1 and 2. Diagnosis of all medical injuries was performed by 2 first-team sports medicine doctors with >15 years of experience in professional football. Imaging of injuries was performed using clinical ultrasound and/or magnetic resonance imaging as determined by the responsible physician. Medical illness was excluded due to not directly depending on external mechanical load factors and therefore is out of the scope of the present study. Upper-extremity and trunk soft tissue injuries were also excluded.

Training and games performed during international duty periods were not considered for analysis; additionally, injuries sustained during those periods were not accounted for the overall time loss and injury incidence reported. Goalkeepers were also not included in the present study as this field position is subject to

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Table 1 Total Soft-Tissue Injuries

Region	Injuries (%)		Onset		Mechanism of injury (n)	Recurrences (n)		Layoff days	
	C	NC	Acute	Gradual		C	NC	C	NC
Medical-attention injuries									
Hip									
Obturator muscle injury	1 (1.5)	0	1	0	Running (1)	—	—	—	—
Pubic-related groin pain	1 (1.5)	0	0	1	Running (1)	—	—	—	—
Groin									
Adductor muscle injury	6 (9.2)	6 (9.2)	3	9	Sprint (1), kicking (1), running (12)	—	—	—	—
Adductor-related groin pain	1 (1.5)	6 (9.2)	0	7	Running (7)	—	—	—	—
Adductor tendinopathy	0	1 (1.5)	0	1	Running (1)	—	—	—	—
Thigh									
Hamstrings muscle injury	6 (9.2)	9 (13.8)	1	14	Sprint (1), running (14)	—	—	—	—
Hamstrings tendinopathy	0	5 (7.7)	0	5	Running (5)	—	—	—	—
Quadriceps muscle injury	0	1 (1.5)	0	1		—	—	—	—
Lower leg									
Calf muscle injury	6 (9.2)	8 (12.3)	3	11	Landing (1), tackle (1), kicking (1), running (11)	—	—	—	—
Achilles tendinopathy	0	8 (12.3)	0	8	Running (8)	—	—	—	—
Time-loss injuries									
Hip									
External obturator	1 (2.6)	0	1	0	Tackle (1)	0	0	15	0
Medium gluteus	1 (2.6)	0	1	0	Running (1)	0	0	17	0
Groin									
Adductor muscle injury	4 (10.8)	1 (2.6)	4	1	Running (2), kicking (1)	0	0	48	3
Adductor-related groin pain	0	1 (2.6)	0	1	Controlling the ball (1)	0	0	0	81
Thigh									
Hamstrings muscle injury	10 (25.6)	10 (25.6)	18	2	Running (16), kicking (2), controlling the ball (1), landing (1)	2	4	194	177
Hamstrings tendinopathy	0	1 (2.6)	1	0	Kicking (1)	0	0	0	16
Quadriceps muscle injury	0	3 (7.7)	3	0	Running (2), kicking (1)	0	0	0	41
Lower leg									
Calf muscle injury	1 (2.6)	4 (10.4)	2	2	Running (4)	0	0	8	69
Achilles tendinopathy	0	2 (5.2)	2	0	Running (2)	0	0	0	43

Abbreviations: C, congested; NC, noncongested.

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Table 2 Time Loss and Soft-Tissue Injuries

Region	Injury incidence (per 1000 h, mean [SD])					Injury burden (days lost per 1000 h, mean [SD])				
	Congested	Noncongested	95% CI	*P*	Effect size	Congested	Noncongested	95% CI	*P*	Effect size
Hip										
External obturator	0.15	0	—	—	—	0.77	0	—	—	—
Medium gluteus	0.15	0	—	—	—	2.63	0	—	—	—
Groin										
Adductor muscle injury	1.03	0.22	−7.80 to 12.96	.201	0.800	13.48	0.65	−23.30 to 207.36	.201	0.800
Adductor-related groin pain	0	0.24	—	—	—	0	19.36	—	—	—
Thigh										
Hamstrings muscle injury	2.60	1.47	−2.40 to 8.30	.196	0.436	52.53	31.29	−137.91 to 226.01	.410	0.282
Hamstrings tendinopathy	0	0.07	—	—	—	0	1.19	—	—	—
Quadriceps muscle injury	0	0.27	—	—	—	0	3.65	—	—	—
Lower leg										
Calf muscle injury	0.32	0.59	−5.00 to 5.00	.371	−1.000	2.59	11.95	−171.02 to 171.02	.371	−1.000
Achilles tendinopathy	0	0.16	—	—	—		6.80	—	—	—
Total	3.9 (8.3)	3.2 (5.4)	−5.9 to 7.1	.781	0.074	71.8 (189.0)	60.5 (123.0)	−109 to 57.9	.668	0.114
Training injuries	0.7 (3.4)	1.9 (4.0)	−6.3 to 7.9	.932	0.033	15.3 (55.4)	47.2 (122.0)	−303 to 46.9	.126	−0.513
Match injuries	9.7 (23.6)	8.9 (22.5)	−10.9 to 2.5	.197	−0.455	48.4 (163.0)	99.0 (289.0)	−596 to 148.2	.255	−0.385

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different training and match physical demands compared to outfield players, and therefore, data monitoring is also specific to this position. All data were anonymized before analysis, and informed consent was obtained from all players participating in this study. Procedures for clinical data processing were performed according to the Declaration of Helsinki as revised in 2013.

Statistical Analysis

All statistical analyses were conducted using Jamovi (version 1.6, the Jamovi Project, 2024). Standard descriptive statistics were used to summarize the data. Data normality was first tested using the Shapiro–Wilk test. A paired samples *t* test was used to compare the differences between congested and noncongested periods. When normality was not observed, the Wilcoxon signed-rank test was conducted. Statistical significance was set at $P < .05$.

Results

A total of 65 medical attention and 39 time-loss soft tissue injuries occurred during the 2 seasons. Time-loss injuries led to 704 layoff days (Table 1) and 113 official matches missed. Layoff days resulting from soft tissue injuries accounted for 52.7% of the total layoff (n = 1337) and 49.8% of total games missed (n = 227) due to injury.

The total soft-tissue injury rates for the 2 seasons were 3.9 and 3.2 injuries per 1000 hours, for the 2 considered periods, respectively, with an average of 3.6 injuries per 1000 hours. Total injury burden was 71.8 and 60.5 days lost per hour for congested and noncongested periods, respectively, with an average of 66.2 days lost per hour of training and competition. No significant differences were observed between periods for total injury rates and injury burden. There was an increase in training injury rate during noncongested periods (1.94 injuries per 1000 h), together with noncongested training and match injury burden (47.2 and 99.0 d lost per 1000 h, respectively), but with no statistical significance.

Total physical load outputs comparison between 2 periods showed significantly higher values per week for noncongested periods ($P < .01$), except for AD3, in which no difference was registered between congested and noncongested periods. Training physical outputs per week were higher during noncongested periods, whereas match physical outputs were significantly higher during congested ones (Table 3).

Discussion

Injury Rate and Injury Burden

This study analyzed the influence of congested and noncongested periods on soft injury rate and burden and external load. The hypothesis of our study was not verified, as no significant effect of a congested schedule was observed for training and match injury rates and burden. The average total match soft tissue injury rate for the 2 periods considered was 9.3 per 1000 hours, and 9.7 and 8.9 per 1000 hours for congested and noncongested periods, respectively. The average total match injury burden was 73.7, and 48.4 and 99.0 days lost per 1000 hours for congested and noncongested

Table 3 External Load During Congested and Noncongested Periods

Variable	Congested (mean [SD])	Noncongested (mean [SD])	95% CI	*P*	Effect size
Total external load					
High-speed running, m	1643 (613)	1968 (559)	−448.14 to −201.689	<.001*	−0.766
Sprint week, m	339 (150)	404 (157)	−95.38 to −34.41	<.001*	−0.618
V80 week, m	101.0 (59.6)	140.0 (73.5)	−45.50 to −20.00	<.001*	−0.745
V90 week, m	12.2 (10.8)	19.0 (15.5)	−10.00 to −3.00	<.001*	−0.726
AD2 week, m	5137 (1492)	6389 (1481)	−1564.12 to −939.42	<.001*	−1.164
AD3 week, n	613 (260)	629 (167)	−87.00 to 70.00	.689	−0.027
Match external load					
High-speed running, m	3910 (1470)	3298 (1618)	174.00 to 944.00	.002*	0.525
Sprint week, m	827 (379)	699 (384)	52.48 to 202.35	.001*	0.494
V80 week, m	280 (183)	212 (147)	26.00 to 87.50	<.001*	0.605
V90 week, m	32.5 (32.9)	18.8 (17.9)	2.50 to 17.50	.006*	0.478
AD2 week, m	10,321 (4026)	8422 (3820)	655.50 to 2579.00	.002*	0.532
AD3 week, n	891 (411)	708 (309)	58.50 to 233.50	.002*	0.515
Training external load					
High-speed running, m	832 (409)	1526 (540)	−823.19 to −564.31	<.001*	−1.556
Sprint week, m	161 (105)	321 (157)	−194.50 to −115.50	<.001*	−0.974
V80 week, m	45.6 (39.3)	63.5 (64.5)	−34.95 to −0.80	.041*	−0.304
V90 week, m	8.13 (12.1)	16.3 (14.4)	−11.00 to −5.50	<.001*	−0.869
AD2 week, m	3426 (1242)	5605 (1374)	−2396.02 to −1962.32	<.001*	−2.918
AD3 week, n	351 (140)	574 (165)	−245.00 to −200.50	<.001*	−1.000

Abbreviations: AD2, sum of distance accelerating and decelerating above 2 m/s^2; AD3, sum of the number of accelerations and decelerations above 3 m/s^2; V80 and V90, relative velocity distance above 80% and 90% of the individual maximum speed, respectively.
*Significant difference ($P < .05$).

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periods. Despite no statistical significance, the congested match injury rate was superior to the training injury rate, but these injuries presented less severity as the match injury burden for the congested periods presented an opposite tendency and was inferior to noncongested periods. These numbers are inferior to what is usually reported in research in this field.[7,8,18] In a study from the UEFA Champions League injury study group, Bengtsson et al[7] reported match muscle injury rates when 4 days separated competitions. Their results showed a range of injury incidence of 8.2 and 11.9 injuries per 1000 hours, averaging 10.9 muscle injuries per 1000 hours of competition. The same research reported 10.6 muscle injuries per 1000 hours of competition for the same period of days between matches for an analysis of 14 consecutive seasons.[8] Pinheiro et al[9] also observed a high muscle injury rate when analyzing 35 consecutive matches of a Brazilian football team during 35 matches in 17 weeks. Their results showed 19.0 injuries per 1000 hours during congested periods, however with no changes in physical and physiological outputs for the same period.

Some methodological differences in injury data collection might exist between the studies from the UEFA Champions League injury study group and the present one. First, whereas the present study reports all soft tissue injuries causing absence from training and competition, particularly the study from Bengtsson et al[7] seemed to be focused on 4 muscle groups, namely, hamstrings, quadriceps, adductors, and calf, being unclear whether other injuries were included. Additionally, from our total soft tissue injury cohort, there was a particular case with a significantly high contribution to the total injury rate and injury burden. A single player sustained 5 injuries during the 2022–2023 season, causing 113 layoff days and 18 games missed, with a correspondent match injury rate for that player of 103.9 soft tissue injuries per 1000 hours, and a burden of 587.5 days lost. This player was unavailable for training and competition for 42% and 43% of the time, respectively, and if it was excluded as an outlier, the soft tissue injury rates would be reduced for a total of 3.1 injuries per 1000 hours, and a total injury burden to 59.5 days lost per 1000 hours.

Match and Training External Load

Total external load metrics displayed higher scores per week during noncongested weeks compared to congested weeks, apart from AD3 (Table 3). Training metrics were significantly higher during noncongested weeks, as a result of higher training exposure and more frequency of intense sessions, and match physical outputs were higher during congested weeks due to higher match exposure. AD3 was significantly higher in congested periods for matches and noncongested periods for training. Therefore, the total amount of meters of AD3 per week was maintained constant, with no statistical significance for total AD3 between the 2 periods observed. Players displayed higher amounts of match AD3 per week when the fixtures were congested or accumulated higher frequency of this metric in training when the opportunities to train with higher intensity and/or more frequently were possible, such as noncongested periods. High accelerations and particularly high decelerations are associated with both a high metabolic cost and particular a high mechanical strain.[15] Independently of the type of drill performed during training, or match tactical and strategic displays, AD3 will usually be performed when football practice occurs with competitive intensity.[19] Despite the design and setup of some training drills may lead to an increase in the density of AD3, this metric will be less dependent on the availability of space to be performed, contrary to the ones related to higher speed zone ones, as the latter will be dependent on the availability of space during a drill to reach higher velocities.

Taking the opportunities during noncongested periods to increase exposure across several training load metrics could have contributed to a load accommodation and may have had a protective effect toward soft tissue injury during congested periods, in which match load increased due to a higher exposure, and therefore, the risk of injury would be also expected to increase,[7,18] which was not observed in this study.

External-Load Data V80 and V90 Relative Thresholds

In this study, higher relative speed zones such as V80 and V90 were analyzed together with HSR and sprint usually adopted in football.[15] The importance of higher speed zones for performance and injury prevention has been referred to in research.[20,21] For muscles such as the hamstrings, from a neuromuscular perspective, maximal sprinting places the highest strain both during the acceleration and maximal velocity phases.[22,23] Targeting for significant volumes (ie, game demands) of V80 during practice might be more challenging than V90. The V90 normally presents lower absolute numbers which are more subject to be impacted with a low number of drills, with the condition that maximal velocity is involved. For example, a velocity-specific exposure drill such as a 40 m maximal sprint will allow the addition of a few meters of V90 and ensure weekly targets are fulfilled. The V80, however, will present higher weekly numbers which will demand players to run at high velocities during training, to achieve weekly targeted volumes, usually using game physical outputs as references. However, when small spaces are used in training players do not have enough area per player to reach 80% of their individual max velocity. To create sufficient volume per week that could differentiate a regular from a congested week, a significantly extra number of meters would have to be achieved in training through drills that specifically promote entrances on V80. Although the same rationale is valid for HSR and sprint distances, these normally constitute lower percentages of the player's maximal speed, and therefore, it is harder to accumulate V80 distances during training than HSR and sprint. Similar V80 distances have been analyzed in the study of Gualtieri et al,[24] in which starters were compared with nonstarter players for 2 congested periods during a football season. Results showed a tendency for moderate to large variations between V80 between players usually starting matches and nonstarters. This comes as a particular point of interest since V80 was the only velocity-related metric presenting differences between groups. Distances covered above 20 and 25 km/h, in what is considered HSR (19.8 km/h) and sprint (25.2 km/h) zones in football presented similar results between groups.[15] Considering the importance of this stimulus from a musculoskeletal perspective, the importance of individualizing very high SR thresholds is stressed by the authors and reinforced by the present study through the monitoring of V80 and V90.

Factors Influencing Injury Rate and Burden

Results for soft tissue injury rates and the burden of the present study might have been affected by several factors. External load associated with training and competition has been identified as one of the primary reasons for soft tissue injury occurrence.[13] The study of Anderson et al[20] suggests that maintaining regular loads throughout all squad players constitutes a challenging task but may be of fundamental importance to maintain preparedness throughout the season across all players and potentially decrease injury risk. Congested calendars dictate that players will be exposed to metrics

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exclusively in accordance with match demands. For some field positions, and depending on team strategy, level, and competition, this might cause an underexposure to specific load parameters, such as sprint or higher speed zones. Compensating the randomness of match load exposure through training during regular periods and leveling different metrics might be useful to (re)adapt players to fixture congestion demands. Despite the importance of external load, other relevant factors might have contributed to the present study results. Among those, the post-COVID increase in the number of substitutions (5 per match) may be one of the most important. It is worthwhile mentioning that previous studies[1,2,5,6] investigating congested versus noncongested fixture periods on injury rate were performed when only 3 substitutions were allowed. Since the introduction of the 5 substitutions rule, more substitutions per match are performed with Garcia-Aliaga et al[25] suggesting that this could potentially be a protection against injuries.

The injury rates presented in the current study will hardly be solely the consequence of external load volumes per week. Factors such as the training periodization adopted, particularly with regard to recovery off pitch and/or day off in the first- and second-day postmatch, respectively, may contribute to potentiate recovery and decrease injury occurrence.[26] Another important factor is player rotation from match to match or number of consecutive matches performed. In the present study, a distinctive analysis for players with higher match exposure, particularly during consecutive games in congested schedules, was not performed. Future analysis should understand the impact of consecutive match participation on soft tissue injury rate and the present load metrics analyzed in this study, differentiating players with higher match exposure from the remaining.

Also, a very important intrinsic factor such as the players' injury record is of relevance when analyzing the present results, as it comes as the number one risk factor for injury risk[21] or extrinsic factors such as appropriate communication between medical and coaching staff could have played a relevant role.[27]

Lower limb strengthening was also performed every time the calendar allowed at least 1 recovery day afterward, and it could have also contributed to the injuries sustained throughout the 2 seasons. Strength training, particularly by promoting significant muscle fiber changes in pennation angles and fascicle lengths, has been previously associated with higher ability from muscles to sustain strains associated with football practice, while promoting limb symmetry in several muscle groups.[28,29]

Practical Applications

- Congested and noncongested periods showed similar soft tissue injury rates and burden.
- External load metrics increased during congested periods for matches, while for training sessions the load metrics were greater during noncongested periods.
- Relative speed zones such as V80 and V90 can be used as an additional metric for match load analysis and training planning, due to the mechanical strain associated, which might favor positive adaptations in football players and work as a protective factor against soft tissue injuries.

Conclusions

The present study described the effect of a congested schedule on soft-tissue injury and external load performed per week for matches and training. No effect of a congested schedule was observed in soft-tissue injury rates. Higher match exposure during congested periods increased external load performed per week, and during noncongested periods, training load was superior to congested weeks. Additionally, relative metrics for speed distances are included that may be useful to individualize and plan training loads, helping to determine squad rotation strategies to achieve more availability and less injury burden due to injury occurrence, while aiming for optimal conditioning levels among players.

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